

A Reader-Friendly Guide to PPG-12

Isolated-Prompt-Photon Cross Section in $p+p$ Collisions at
 $\sqrt{s} = 200$ GeV with sPHENIX

An explanatory companion to sPHENIX note sPH-ppg-2024-012 (v4)

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Explainer prepared for an HEP graduate reader

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Contents

1	Why measure prompt photons? (Physics motivation)	1
1.1	What is a “prompt photon”?	1
1.2	Why physicists care	2
1.3	The isolation idea	2
2	The detector and the data	2
2.1	The sPHENIX calorimeters in one minute	2
2.2	Data and Monte Carlo	3
2.3	Pileup: “double interactions”	3
2.4	Trigger and event selection	3
3	Finding the photons (reconstruction & identification)	4
3.1	From towers to clusters	4
3.2	Energy scale and resolution	4
3.3	The background problem	4
3.4	Killing the beam junk (the NPB BDT)	4
3.5	Picking out photons (the photon-ID BDT)	5
3.6	The isolation variable in the detector	5
4	Extracting the cross section (analysis procedure)	5
4.1	The workflow at a glance	5
4.2	Purity: the ABCD (double-sideband) method	5
4.3	Efficiency	6
4.4	Unfolding	6
4.5	Luminosity	7
5	Systematic uncertainties	7

6 Results	8
6.1 Comparison to PHENIX (an instructive cross-check)	8
7 Selected deep-dives (the appendices, distilled)	8
7.1 Double-interaction (pileup) physics	8
7.2 Other cross-checks worth knowing	8
8 Glossary & takeaways	9

1 Why measure prompt photons? (Physics motivation)

1.1 What is a “prompt photon”?

A **prompt photon** is a real, high-energy photon that emerges from the hard scattering of two partons, as opposed to a photon from the decay of a hadron produced later. There are two kinds:

- **Direct photons:** produced in the elementary $2 \rightarrow 2$ hard scattering itself. At leading order (LO) the two dominant channels are *quark-gluon Compton scattering* $qg \rightarrow q\gamma$ and *quark-antiquark annihilation* $q\bar{q} \rightarrow g\gamma$.
- **Fragmentation photons:** radiated collinearly off a final-state quark as it fragments (essentially bremsstrahlung from a hard parton). These tend to sit *inside* a jet.

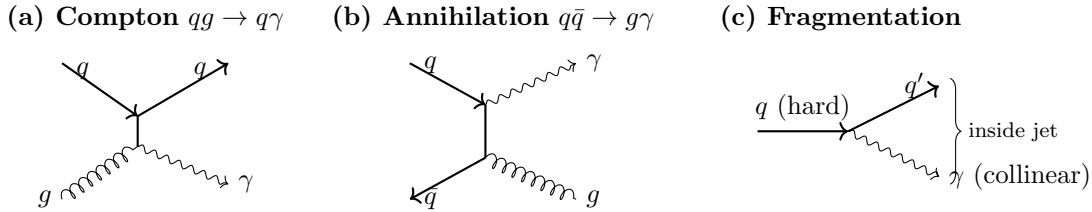


Figure 1: Schematic LO mechanisms (redrawn after Fig. 1 of the note). (a) and (b) make *direct* photons; (c) makes *fragmentation* photons that accompany a jet. At NLO, direct production also receives bremsstrahlung corrections.

1.2 Why physicists care

1. **A stringent test of pQCD.** The cross section can be computed at next-to-leading order (NLO). Comparing data to theory tests our understanding of the strong interaction.
2. **Direct sensitivity to the gluon PDF.** In this kinematic range the Compton process $qg \rightarrow q\gamma$ *dominates*, so the rate is directly proportional to the gluon parton distribution function (PDF) of the proton. RHIC’s $\sqrt{s} = 200$ GeV probes the gluon at parton momentum fractions

$$x \sim \frac{2 E_T^\gamma}{\sqrt{s}} \approx 0.1\text{--}0.3,$$

a *higher- x* window than the LHC reaches — so the two machines are complementary.

3. **A reference for heavy-ion physics.** The $p+p$ result is the baseline against which future sPHENIX measurements in Au+Au (where photons probe the quark–gluon plasma) will be normalized.
4. **Continuity with PHENIX.** It connects directly to the earlier PHENIX direct-photon measurement at the same energy.

1.3 The isolation idea

Fragmentation photons live inside jets, so they are surrounded by other particles. Direct photons are “lonely.” We exploit this with an **isolation** requirement: we sum the transverse energy of everything within a cone of radius $\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2} = 0.3$ around the photon (excluding the photon itself) and demand it be small. At truth level the fiducial definition is

$$E_T^{\text{iso,truth}} < 4 \text{ GeV} \quad \text{in a cone } \Delta R = 0.3$$

The note shows (its Fig. 2–3) that at a 4 GeV threshold, *direct* photons stay essentially 100% efficient while *fragmentation* photons are strongly suppressed. Isolation also rejects the dominant background — π^0/η decays inside jets.

2 The detector and the data

2.1 The sPHENIX calorimeters in one minute

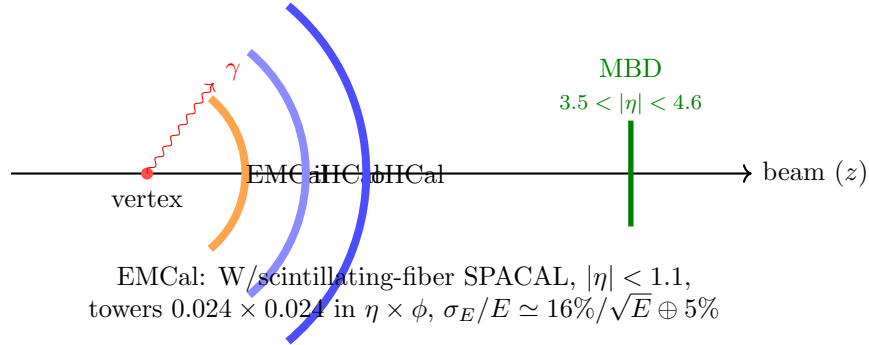


Figure 2: Schematic of the relevant sPHENIX subsystems (not to scale). The EMCal measures the photon energy; the two HCal layers contain hadronic showers and contribute to isolation; the forward MBD provides the trigger and the collision vertex.

- **EMCal** (electromagnetic calorimeter): a tungsten / scintillating-fiber SPACAL detector, full 2π in azimuth, $|\eta| < 1.1$, finely segmented into 96×256 towers ($\Delta\eta \times \Delta\phi \approx 0.024 \times 0.024$). Single-photon resolution $\sigma_E/E \simeq 16\%/\sqrt{E[\text{GeV}]} \oplus 5\%$. *This is where photons are measured.*
- **iHCal + oHCal** (inner/outer hadron calorimeters): sit behind the EMCal; catch hadronic energy and feed the isolation sum.
- **MBD** (Minimum Bias Detector): two arrays of 64 quartz-Cherenkov PMTs at $3.5 < |\eta| < 4.6$. Provides the minimum-bias trigger and reconstructs the collision vertex z .

2.2 Data and Monte Carlo

- **Data:** Run-24 $p+p$ at $\sqrt{s} = 200$ GeV, runs 47289–53880, integrated luminosity $\mathcal{L} = 64.37 \text{ pb}^{-1}$ (production `ana521`). Selected from a centrally produced calorimeter “golden run” list with basic quality cuts.
- **MC:** PYTHIA-8 (Detroit tune) hard-QCD + prompt-photon events passed through GEANT-4 with the full sPHENIX geometry. “Photon” samples (signal-enriched) and “jet” samples (background) are generated at several minimum- p_T thresholds (5/10/20 GeV for photons; 8/12/20/30/40 GeV for jets) to keep statistics high at large p_T .

2.3 Pileup: “double interactions”

At RHIC two separate $p+p$ collisions can occur in the same bunch crossing (**pileup**, here called a **double interaction**, DI). The MBD then reconstructs a *single* vertex at the average of the two collision points, which distorts the inferred photon η and the shower shapes. To handle this honestly, a set of full-GEANT *DI MC* samples (two overlapping collisions) is blended with the single-interaction (SI) samples. The data split into two crossing-angle periods:

Table 1: The two running periods (see Sec. 7.1 for the DI machinery).

Period	Runs	$\mathcal{L}$	lumi region σ_z	cluster-weighted DI
0 mrad	47289–51274	47 pb^{-1}	$\simeq 54 \text{ cm}$	$\sim 22.4\%$
1.5 mrad	51274–54000	17 pb^{-1}	$\simeq 22 \text{ cm}$	$\sim 7.9\%$

The 1.5 mrad crossing angle geometrically shrinks the luminous region, which reduces pileup.

2.4 Trigger and event selection

Events are taken with the “**Photon 4 GeV**” Level-1 trigger (a high-energy EMCal cluster in coincidence with the MBD). The trigger efficiency is measured bin-by-bin against an unbiased MBD-only reference,

$$\varepsilon_{\text{trig}}(E_T) = \frac{N(\text{scaled}[10] \ \& \ \text{live}[\text{Nbit}])}{N(\text{scaled}[10])}, \quad (1)$$

and fit with a saturating (Gumbel) curve. The trigger is essentially fully efficient ($\gtrsim 99.7\%$) above $E_T^\gamma \approx 12 \text{ GeV}$, which is why the result starts there. Events must have a reconstructed vertex $|z_{\text{vtx}}^{\text{MBD}}| < 60 \text{ cm}$.

3 Finding the photons (reconstruction & identification)

3.1 From towers to clusters

EMCal towers above $\sim 70 \text{ MeV}$ that share an edge are merged into **clusters** (`RawClusterBuilderTemplate`), with a splitting step for overlapping showers. The cluster energy is the tower sum; its (η, ϕ) are computed relative to the MBD vertex.

3.2 Energy scale and resolution

The detector response is calibrated in MC by comparing reconstructed cluster E_T to the truth photon E_T . In each truth- E_T bin the ratio $E_T^{\text{reco}}/E_T^{\text{truth}}$ is fit with a Gaussian: the mean (≈ 0.99 , the energy *scale*) and the width ($\approx 5\text{--}7\%$, the *resolution*) are both excellent. The small residual scale/resolution effects are absorbed by unfolding (Sec. 4.4).

3.3 The background problem

We separate signal from background using **shower shapes** — moments and energy ratios of the towers in a cluster. A genuine single photon makes a narrow, symmetric shower with most energy in the central tower; a merged π^0 is broader or more lopsided. Key variables include:

Table 2: The most important shower-shape variables (selected from the note’s Table 4).

Symbol	Meaning
$E_{1\times 1}/E$	central-tower energy fraction (peaked for photons)
$w_\eta^{\text{COGX}}, w_\phi^{\text{COGX}}$	energy-weighted second moments (shower width) in η, ϕ
$E_{3\times 2}/E_{3\times 5}$	energy in a 3×2 vs. 3×5 tower window
R_{had}	$E_T^{\text{HCal}}/(E_T^{\text{EMCal}} + E_T^{\text{HCal}})$ (hadronic leakage)
$\text{Et1} - \text{Et4}$	energies/asymmetries of the central 2×2 tower block

3.4 Killing the beam junk (the NPB BDT)

NPB clusters are spread out in η (large w_η^{COGX}) and arrive at anomalous times relative to the MBD. A first cut $w_\eta^{\text{COGX}} > 1.0$ removes the worst streaks; the rest is handled by an **XGBoost boosted decision tree (BDT)** trained to separate real collision clusters (from MC) from streak clusters (tagged in data by anomalous timing $t_{\text{cluster}} - t_{\text{MBD}} < -7$ ns). The BDT uses 25 features (note: *not* timing, so timing stays available as an independent validation). The purity is measured with a data-driven template fit using the cluster–MBD time tail. The nominal cut $\text{NPB} > 0.5$ gives $\sim 99\%$ purity while keeping $\sim 99\%$ of real clusters.

3.5 Picking out photons (the photon-ID BDT)

A *second* XGBoost BDT combines 11 shower-shape features into a single photon-identification discriminant. Clusters first pass a rectangular **pre-selection** ($E_{1\times 1}/E_{3\times 3} < 0.98$, $0.6 < \text{Et1} < 1.0$, $0.8 < E_{3\times 2}/E_{3\times 5} < 1.0$, $\text{NPB} > 0.5$). Then:

- **Tight** = pre-selection + high BDT score ($> 0.8156 - 0.00156 E_T^\gamma$): signal-rich.
- **Non-tight** = pre-selection + an intermediate BDT band: background-rich *control sample*.

3.6 The isolation variable in the detector

At reco level the analysis sums *topological clusters* (a seed-and-grow calorimeter clustering, the same scheme as ATLAS) inside a $\Delta R = 0.4$ cone, subtracting the photon's own energy:

$$E_T^{\text{iso}} = \sum_{\text{topo-clusters in cone}} E_T - E_T^\gamma.$$

Five candidate definitions were compared with ROC curves; the topo-cluster $R = 0.4$ definition won because it is robust against the EMCal noise that drifts upward over the run period (a fixed per-tower threshold sum drifts; topo-clustering does not). The cut is set to 80% signal efficiency:

$$E_T^{\text{iso}} < 0.490 + 0.0370 E_T^{\text{reco}} [\text{GeV}].$$

Note the reco isolation uses $R = 0.4$ for detector performance, while the *fiducial* cross section is defined with the truth $R = 0.3$, $E_T^{\text{iso,truth}} < 4 \text{ GeV}$ cut (matching JETPHOX); the isolation *efficiency* converts between the two.

4 Extracting the cross section (analysis procedure)

4.1 The workflow at a glance

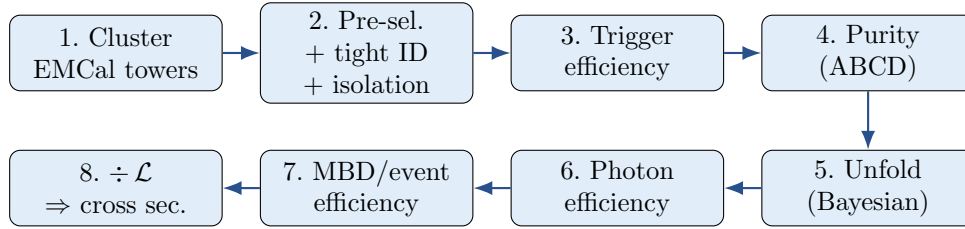


Figure 3: The eight-step analysis chain. Each box is unpacked in the text below.

4.2 Purity: the ABCD (double-sideband) method

Even after tight ID + isolation, some background survives. We must measure the **purity** P = (signal fraction in the signal region) and correct for it. This is done *from data*, using the assumption that the photon-ID and isolation discriminants are approximately **independent** for background.

If background ID and isolation are uncorrelated, then $N_A^{\text{bkg}}/N_B^{\text{bkg}} = N_C^{\text{bkg}}/N_D^{\text{bkg}}$, which lets us *predict* the background in A from the sidebands. The number of signal clusters in A is

$$N_{\text{signal}}^{A,\text{data}} = N_{\text{raw}}^{A,\text{data}} - \frac{N_{\text{raw}}^{B,\text{data}} N_{\text{raw}}^{C,\text{data}}}{N_{\text{raw}}^{D,\text{data}}}. \quad (2)$$

A real subtlety: *signal* photons also leak into the control regions B, C, D. This is corrected using signal-MC leakage fractions $f^{X,\text{MC}} = N_{\text{signal}}^{X,\text{MC}}/N_{\text{signal}}^{A,\text{MC}}$, giving the leakage-corrected equation (the note's Eq. 4). The purity is then

$$P = \frac{N_{\text{signal}}^A}{N^A}. \quad (3)$$

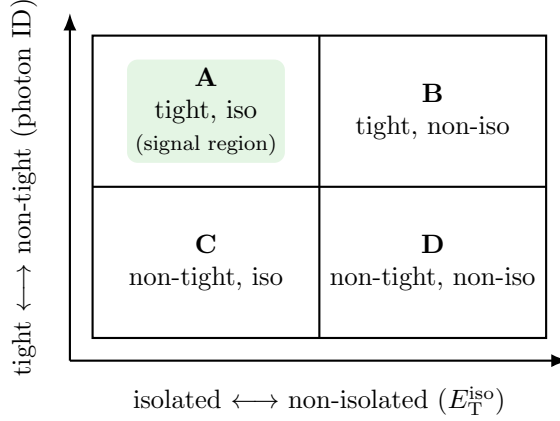


Figure 4: The ABCD method (redrawn after the note’s Fig. 26). Region A is the signal region. If isolation and ID are independent for background, the background in A can be predicted from the three control regions B, C, D.

The measured purity (leakage-corrected) rises with E_T^γ :

$$P \approx 0.51 \text{ (10–12 GeV)} \rightarrow 0.72 \text{ (18–20 GeV)} \rightarrow 0.90 \text{ (> 26 GeV)},$$

exactly as expected since the π^0 background falls faster than the signal.

4.3 Efficiency

The chain of efficiencies, all vs. truth E_T^γ , accounts for everything we lose: **reconstruction** (truth photon $\rightarrow$ matched cluster), **identification** (passes tight ID), **isolation** (passes iso cut), and the **MBD vertex** efficiency (event has an N–S coincidence and a reconstructed vertex in $|z| < 60$ cm). Truth–reco matching uses the GEANT track-ID via `CaloRawClusterEval`. Each is applied as a bin-by-bin correction after unfolding.

4.4 Unfolding

The note uses the **D’Agostini iterative Bayesian** method (`RooUnfold`). Because the EMCal resolution is excellent, R is nearly diagonal and unfolding barely moves the spectrum (2–10% effect). The MC prior is reweighted to match the data shape (a Padé fit of the data/MC ratio), and the number of iterations is set to $N_{\text{iter}} = 2$ — the point where iteration-to-iteration changes drop below the statistical error. **Closure tests** (unfold the MC with a matrix built from the same or an independent half of the MC) reproduce the truth spectrum to within $\sim 2\%$.

4.5 Luminosity

The integrated luminosity $\mathcal{L} = 64.37 \text{ pb}^{-1}$ is the beam-delivered value sampled by the Photon-4-GeV trigger, computed from the live MBD coincidence rate divided by the trigger prescale and the calibrated MBD inelastic cross section $\sigma_{\text{MBD}} = 25.2_{-1.7}^{+2.3} \text{ mb}$ (measured in a dedicated van-der-Meer scan). The σ_{MBD} uncertainty gives the asymmetric luminosity systematic $+9.13\% / -6.75\%$.

5 Systematic uncertainties

Every systematic is evaluated by *rerunning the whole analysis* with one ingredient varied and taking the per-bin change in the cross section. The contributions are added in quadrature per E_T^γ bin. Table 3 lists the maximum size of each source.

Table 3: Maximum per-bin systematic by source over $12 < E_T^\gamma < 32$ GeV (from the note’s Table 7). “low/high” are the down/up variations.

Source	low max [%]	high max [%]
Energy scale (flat $\pm 1.1\%$)	10.5	11.9
Energy scale (non-linearity)	7.3	7.3
Energy resolution	21.2	21.2
Purity – tight ID	6.7	6.7
Purity – non-tight ID	2.4	2.4
Purity – non-isolation gap	3.8	1.0
Purity – fit functional form	0.3	0.3
Purity – fit statistics (1σ)	4.4	4.4
Purity – MC closure correction	5.8	5.8
Efficiency – E_T^{iso} shape	4.7	4.7
Efficiency – generator (HERWIG)	5.8	5.8
MBD vertex efficiency	6.0	6.0
Unfolding – response reweight	0.4	0.4
Unfolding – iteration scan	5.5	5.5
Double-interaction	1.6	1.6
Non-physics-background cut	1.0	4.5
Luminosity (flat)	6.8	9.1
Total (per-bin)	30.8	31.7

6 Results

The differential cross section is built as

$$Y_{\text{rec}}(E_T^\gamma) = \text{Unfold} \left[N_{\text{tight,iso}}^\gamma(E_T^\gamma) \times P(E_T^\gamma) \right], \quad (4)$$

$$\frac{d\sigma}{dE_T^\gamma d\eta^\gamma} = \frac{1}{\mathcal{L}} \frac{Y_{\text{rec}}}{\mathcal{E}} \frac{1}{\Delta E_T^\gamma \Delta \eta^\gamma}. \quad (5)$$

That is: raw tight+isolated yield $\rightarrow$ multiply by purity $\rightarrow$ unfold $\rightarrow$ divide by efficiency $\rightarrow$ divide by luminosity and bin widths.

6.1 Comparison to PHENIX (an instructive cross-check)

PHENIX reported *inclusive* direct photons in $|\eta| < 0.25$ as $d\sigma/(d\eta dE_T^\gamma)$. To overlay it fairly on the sPHENIX $|\eta| < 0.7$ isolated measurement, two corrections are applied:

1. **Bin-width recovery:** a smooth modified-power-law fit removes the few-percent distortion of plotting a steeply falling spectrum at the bin centre.
2. **Pseudorapidity-density rescaling:** the ratio $R(E_T^\gamma) = (dN/d\eta)_{|\eta|<0.25}/(dN/d\eta)_{|\eta|<0.7}$ from truth MC. It is ≈ 1 below 14 GeV and rises to ≈ 1.22 at the highest E_T^γ , because higher- x production is more central in rapidity.

After these, the corrected PHENIX spectrum agrees with sPHENIX across the overlap.

7 Selected deep-dives (the appendices, distilled)

The note’s appendices are cross-checks that build confidence in the result. The most physically interesting ones:

7.1 Double-interaction (pileup) physics

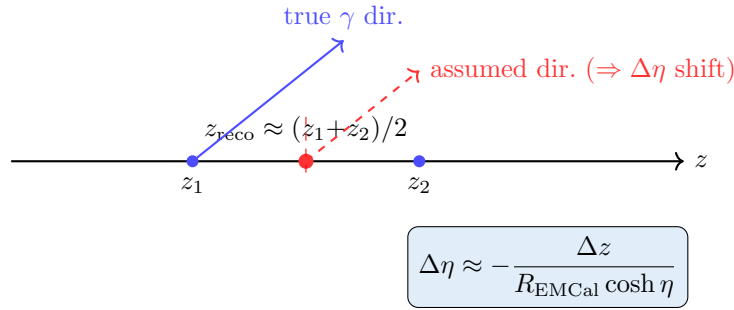


Figure 5: The vertex-shift mechanism. Two collisions at z_1, z_2 give one reconstructed vertex at their average. A photon born at z_1 but assigned to z_{reco} acquires a wrong η , which distorts its shower shapes and feeds the E_T^γ -dependent BDT threshold.

The mean number of interactions per crossing μ is derived from the MBD rate; a zero-truncated Poisson gives the single/double fractions, converted to *cluster-weighted* mixing weights $w_{\text{single}}, w_{\text{double}}$ (Table 4). The dominant DI effect is on **reconstruction efficiency** (a ~ 20 percentage-point drop in pure-DI MC, from cluster loss + shifted kinematics) and on the **photon-ID BDT** (a ~ 13 pp drop). Crucially, all of this is absorbed *automatically* because the efficiency, response matrix, and background template are all built from the SI+DI *blended* MC — there is no separate ad-hoc DI correction. A truth-vertex reweight (factorized $w(z_h)w(z_{\text{mb}})$ for DI events) further aligns the MC vertex distribution to data.

7.2 Other cross-checks worth knowing

- **Back-to-back jet test of the NPB BDT:** in a sample physically depleted of beam junk (require a recoil jet), turning the NPB BDT on/off changes the cross section by $\leq 1\%$ — proving the BDT does not bias real photons.

Table 4: Pileup fractions and mixing weights (the note’s Table 10). f_1, f_2 are event-level single/double fractions; w are cluster-weighted.

Crossing	μ_{corr}	f_1	f_2	w_{single}	w_{double}
0 mrad	0.24	0.879	0.111	0.776	0.224
1.5 mrad	0.08	0.960	0.039	0.921	0.079

- **ϕ -symmetry dead-tower study:** because the cluster rate should be flat in ϕ at fixed η , deviations flag dead towers not modelled in MC. Masking them symmetrically in data and MC shifts the integrated cross section by $\sim 2\%$ (assigned as an acceptance systematic).
- **jetphox PDF comparison:** swapping CT14nlo $\rightarrow$ CT14lo $\rightarrow$ NNPDF3.1 shows the LO-vs-NLO gluon difference (a crossover near $E_T^\gamma \approx 14$ GeV); the two NLO PDF families agree to within ± 3 –20%.
- **herwig generator cross-check:** re-running with HERWIG-7 instead of PYTHIA-8 mainly shifts the isolation efficiency by 3–6% (HERWIG has less underlying-event activity); propagated as an efficiency systematic.

8 Glossary & takeaways

Table 5: Quick-reference glossary.

Term	Meaning
Prompt photon	photon from the hard scatter (direct or fragmentation)
Direct / fragmentation γ	from $2 \rightarrow 2$ vertex / radiated off a final-state quark
Isolation	small extra energy in a cone $\Rightarrow$ “lonely” photon $\Rightarrow$ signal
Shower shape	tower-level moments distinguishing γ from π^0 /junk
NPB	non-physics background (beam-gas/halo streaks)
BDT	boosted decision tree (multivariate classifier; XGBoost here)
ABCD method	data-driven background subtraction via 4 regions
Purity P	signal fraction in the tight+isolated region
Unfolding	undo detector smearing to recover the true spectrum
DI / pileup	two $p+p$ collisions in one bunch crossing
PDF	parton distribution function (here, the gluon is key)

Note: the official note flags its ECal energy-scale variation ($\pm 1.1\%$) as a placeholder pending the Calorimeter Calibration group’s final guidance, so the high- E_T^γ systematic envelope may be re-evaluated.

Key references (from the note): PHENIX direct photons, PRD **86**, 072008 (2012); PYTHIA-8, Comput. Phys. Commun. **191**, 159 (2015); GEANT-4, NIM A **506**, 250 (2003); D’Agostini unfolding, arXiv:1010.0632; ATLAS topo-clusters, EPJC **77**, 490 (2017); Detroit tune, PRD **105**, 016011 (2022); BFG fragmentation functions, EPJ **2**, 529 (1998).